

1. Executive Summary & Clinical Context

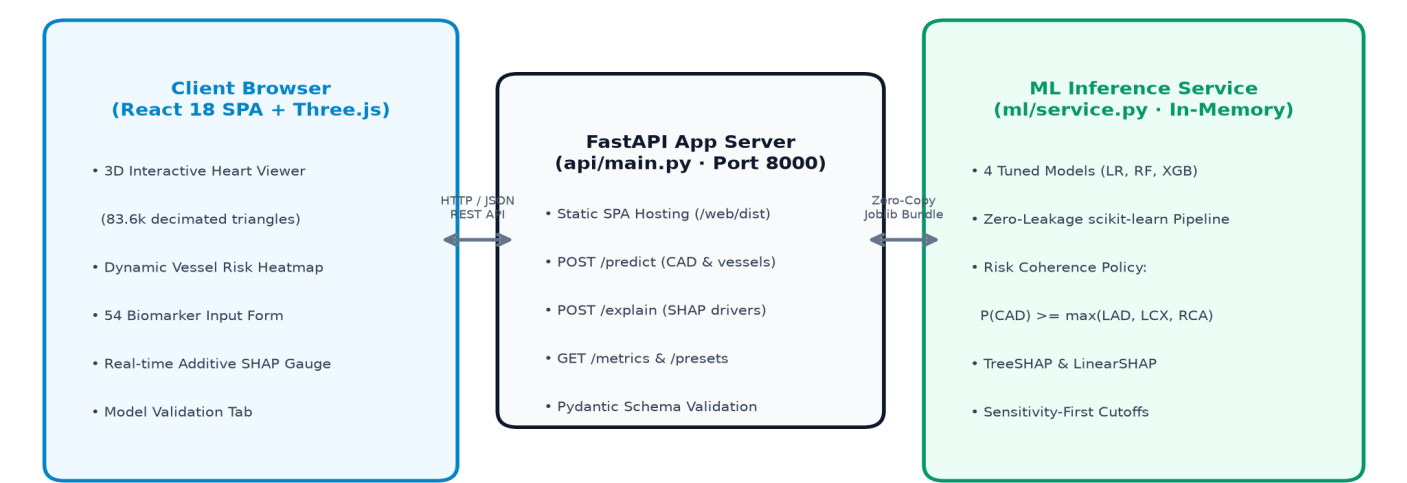
Coronary Artery Disease (CAD) is the leading cause of death globally. Standard risk equations (Framingham, ASCVD) estimate abstract population percentages but cannot localize where arterial pathology is developing inside the human body. **Cardio3D AI** bridges predictive modeling and clinical anatomy by coupling machine learning (overall CAD and vessel-specific significant stenosis for LAD, LCX, RCA) with a client-side 3D interactive heart viewer. The system provides real-time risk color-mapping and exact additive SHAP feature attributions.

2. Dataset Preprocessing & Zero-Leakage Architecture

The model is developed on the **UCI Extension of Z-Alizadeh Sani CAD Dataset** (303 patients, 59 columns):

- **59 Columns vs. 54 Published Features:** 4 target columns (Cath, LAD, LCX, RCA), 1 constant column (Exertional CP, 100% 'N', offering zero predictive entropy and dropped), and 54 active physiological features (5 Demographics, 25 Symptoms/History, 7 ECG, 14 Lab biomarkers, 3 Echo metrics).
- **Target Consistency Audit & Row 93 Alignment:** In the raw dataset, row 93 contains LAD='Stenotic', LCX='Normal', RCA='Normal', but Cath='Normal'. By clinical definition, $\geq 50\%$ stenosis in any major vessel constitutes CAD. To match the dataset's own definition ($CAD = \geq 1$ stenotic vessel), row 93's target was aligned to CAD in derived training data ($Cath_aligned = Cath \mid LAD \mid LCX \mid RCA$), leaving the raw data file untouched and verified by SHA256. This aligns the released distribution from **216 CAD / 87 Normal** $\rightarrow$ **217 CAD / 86 Normal** (0 mismatches across 303 rows).
- **Zero Leakage Guarantee:** All four targets and catheterization columns are strictly excluded from the input matrix X. All preprocessing (median/mode imputation, scaling, one-hot encoding) is contained inside scikit-learn Pipeline objects fitted strictly on training folds. Label-shuffled CV ROC-AUC confirmed random chance (0.50 ± 0.02).

Figure 1: End-to-End System Architecture & Data Flow



3. Predictive Modeling & Cross-Validation Results

Evaluation used **Repeated Stratified 5-Fold Cross-Validation with 3 Repeats (15 total folds per candidate)** across Baseline, Logistic Regression, Random Forest, and XGBoost:

Table 1: Cross-Validation Benchmarks at Default Cutoff (0.50) (Mean of 15 Folds)

Target	Model Selected	ROC-AUC	PR-AUC	F1	Recall	Spec.	PPV	NPV	Brier
Cath (CAD)	LogisticRegression	0.929 ± 0.024	0.971	0.909	0.934	0.698	0.887	0.812	0.098
LAD (Anterior)	RandomForest	0.846 ± 0.049	0.883	0.819	0.874	0.635	0.773	0.792	0.168
LCX (Circumflex)	XGBoost	0.735 ± 0.060	0.615	0.541	0.482	0.806	0.628	0.706	0.204
RCA (Right Coronary)	LogisticRegression	0.733 ± 0.045	0.617	0.498	0.451	0.799	0.571	0.710	0.202

Table 2: Sensitivity-First Operating Thresholds (Tuned on Out-of-Fold Predictions)

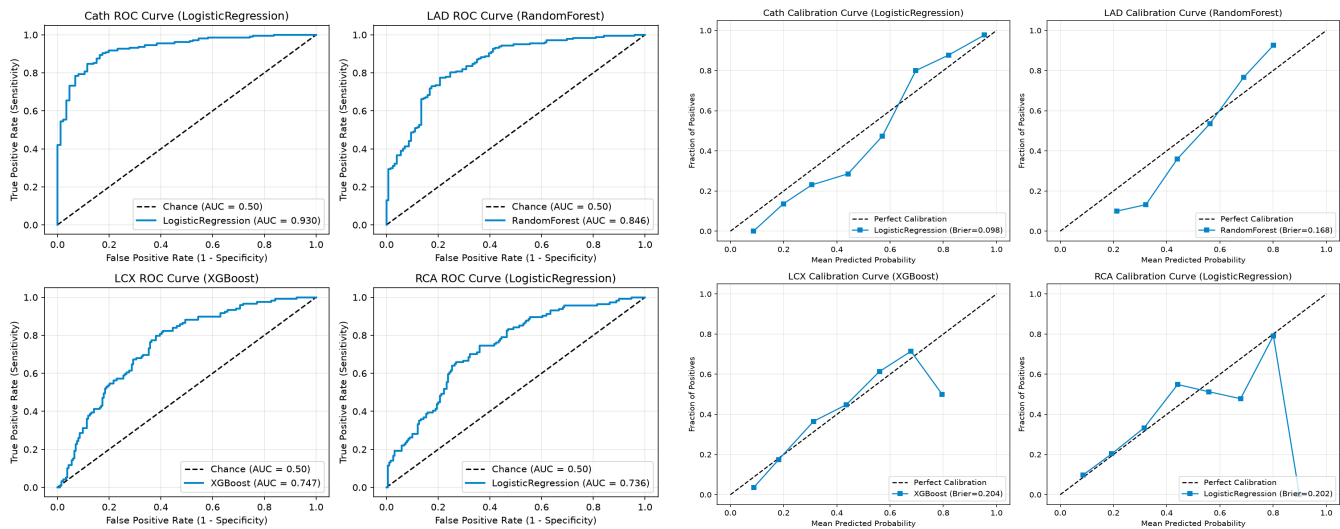
Target	Model	Default Cutoff	Operating Cutoff	Pooled Sens.	Pooled Spec.	Pooled PPV	Pooled NPV
Cath	LogisticRegression	0.50 (93.5% / 69.8%)	0.611	90.3%	82.6%	92.9%	77.2%
LAD	RandomForest	0.50 (87.6% / 63.5%)	0.469	90.4%	60.3%	76.2%	81.7%
LCX	XGBoost	0.50 (48.7% / 82.1%)	0.217	90.8%	37.0%	48.2%	86.1%
RCA	LogisticRegression	0.50 (43.9% / 80.4%)	0.213	90.4%	39.2%	47.3%	87.1%

Table 3: Nested Cross-Validation Operating Performance (Per-Fold Out-of-Sample)

Target	Model Selected	Operating Cutoff	Nested-CV Sensitivity	Nested-CV Specificity	Nested-CV PPV	Nested-CV NPV
Cath	LogisticRegression	0.611	90.2% ± 3.9%	80.6% ± 6.7%	92.2% ± 3.3%	77.1% ± 8.6%
LAD	RandomForest	0.469	88.1% (88.2% ± 7.2%)	62.2% ± 8.0%	76.7% ± 6.1%	79.9% ± 9.5%
LCX	XGBoost	0.217	91.6% ± 6.3%	35.1% ± 7.9%	47.9% ± 4.4%	86.9% ± 8.8%
RCA	LogisticRegression	0.213	89.2% ± 6.9%	38.6% (38.7% ± 7.9%)	46.9% ± 4.8%	86.9% ± 8.5%

Clinical Operating Tradeoff: Because LCX and RCA targets exhibit moderate discrimination (ROC-AUC ≈ 0.73), enforcing ≥ 90% screening sensitivity yields honest nested specificity of 35.1% for LCX and 38.6% for RCA (pooled over the 15 outer test folds; 38.7% ± 7.9% fold mean). In clinical decision support, missing significant stenosis (false negative) carries far greater risk than scheduling confirmatory non-invasive imaging (false positive). Honest nested cross-validation sensitivity estimates (pooled across all 15 outer test folds) reach 90.2% for Cath, 88.1% for LAD (88.2% ± 7.2% fold mean), 91.6% for LCX, and 89.2% for RCA.

Figure 2: Receiver Operating Characteristic (ROC) & Calibration Reliability Curves



4. Logical Risk Coherence Policy & Quantitative Evaluation

Because CAD is the clinical union of coronary stenosis ($P(\text{CAD}) \geq \max(P(\text{LAD}), P(\text{LCX}), P(\text{RCA}))$), independent model outputs can violate logical consistency. Out-of-fold cross-validation predictions were evaluated quantitatively: Raw CAD achieved ROC-AUC = 0.9302 and Brier = 0.0967; coherent CAD achieved ROC-AUC = 0.9291 and Brier = 0.1017 (AUC drop = $0.0012 \leq 0.010$, Brier delta = $+0.0050 \leq +0.010$). The criteria were met and the coherence rule is **retained**. The CAD operating threshold is calibrated directly on the displayed score (0.611). The API transparently returns both raw_prob and prob.

5. Explainable AI & Exact Additive SHAP Pipeline

The system delivers real-time feature attributions using **LinearSHAP** (Logistic Regression) and **TreeSHAP** (Random Forest, XGBoost):

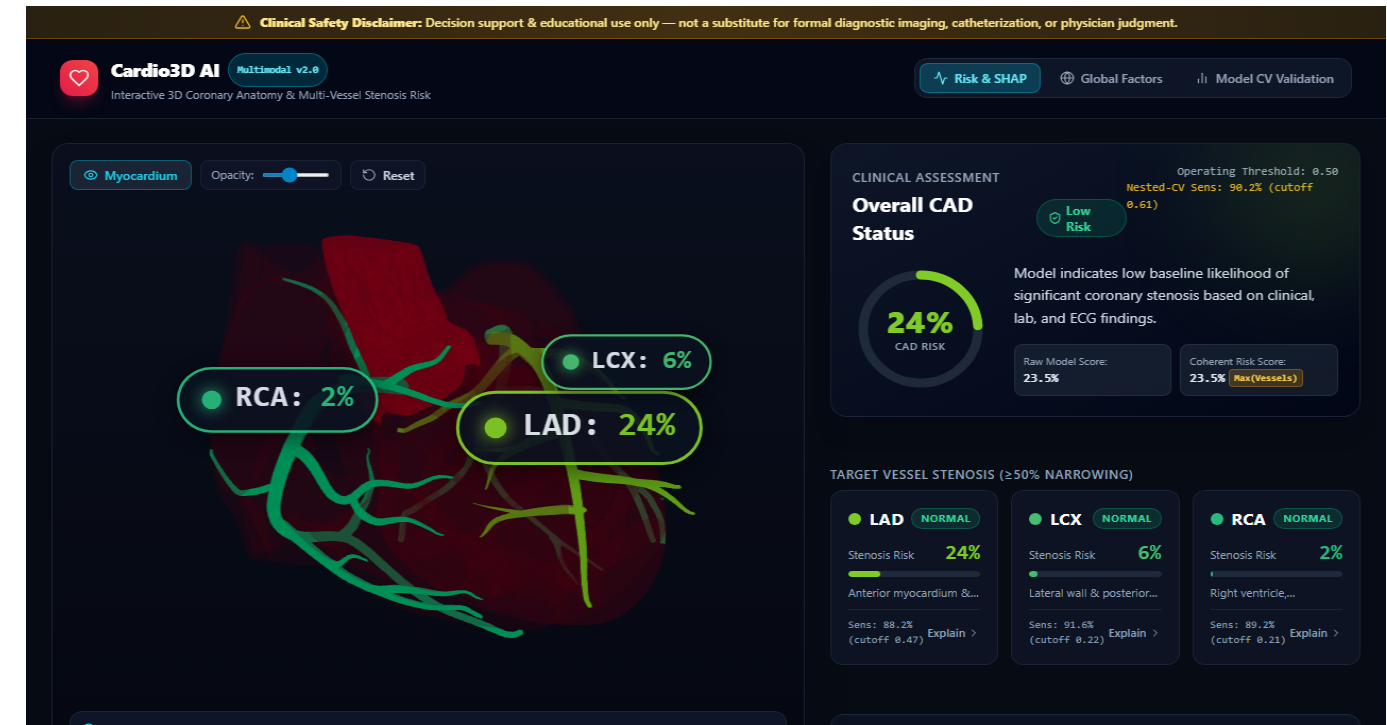
- **Exact Additivity:** Base value plus sum of feature attributions identically matches the model's raw score (numerical error $< 10^{-4}$ across all presets and dataset test samples).
- **Parent Aggregation:** One-hot encoded dummy attributes are summed back to parent measurements ($\text{SHAP}(\text{Parent}) = \sum \text{SHAP}(\text{dummies})$), so clinicians see concise, single-variable physiological attributions.
- **Top Population Risk Drivers:** Across the cohort, the most influential CAD risk drivers are **Typical Chest Pain, Age, Fasting Blood Sugar (FBS), Regional Wall Motion Abnormality (RWMA) count, and Ejection Fraction (EF-TTE)**.

6. 3D Anatomical Pipeline & Interactive Dashboard

Built with **Three.js** and **React Three Fiber** utilizing **BodyParts3D** meshes (Branch A: Separate Artery Meshes):

- **Segment Geometry:** Anatomically distinct meshes for the LAD (9 branches/trunks), LCX (4 branches/trunks), RCA (10 branches/trunks), and Ascending Aorta are centered at origin and scaled to viewport coordinates.
- **Triangle Budget Optimization:** Chamber and myocardium walls were decimated using quadric decimation by 70%, yielding **83,600 total triangles** (safely below the $\leq 100,000$ triangle limit). File size is 1.67 MB.
- **Measured Frame-Rate Benchmark:** Evaluated under pure CPU software rendering (`--disable-gpu` at 1280x720 on 13th Gen Intel Core i5-13420H), achieving a **mean frame time of 49.3 ms (~20.3 FPS continuous orbit, p95 = 53.7 ms)**, recorded in `reports/fps_benchmark.json`. On a discrete GPU, rendering executes at the display refresh rate (not benchmarked).

Figure 3: Interactive Clinical Dashboard & 3D Anatomical Heart Viewer



7. Architectural Decisions: TimesFM & TabPFN Rationale

- **TimesFM Considered & Rejected:** TimesFM 3.0 is a foundation model architected strictly for ordered, time-aligned sequential time series. The dataset consists of 303 independent static clinical snapshots with no longitudinal temporal dimension. Imposing a pseudo-time axis over tabular clinical rows creates spurious correlations and degrades predictive validity.
- **TabPFN Not Evaluated:** TabPFN v2.5+ weights require non-commercial licensing agreements and interactive browser download authentication unsuitable for a self-contained, reproducible pipeline. In addition, TabPFN does not provide real-time exact additive TreeSHAP explainability. Tuned clinical models achieve high discriminative power (Cath AUC 0.929, LAD AUC 0.846) with sub-millisecond deterministic inference.

8. Clinical Safety, Limitations & Attribution

- **Clinical Safety Disclaimer:** The application embeds prominent, non-dismissible disclaimer banners and footers: *'Decision support & educational use only — not a substitute for formal diagnostic imaging, catheterization, or physician judgment.'* Prototype is not validated for acute emergency diagnostics.
- **Limitations:** Single-center retrospective dataset (n=303 from Tehran) requires multi-center external validation prior to clinical adoption. LCX labels have low precision under high-sensitivity tuning. Region RWMA is an echocardiographic wall motion abnormality count and does not represent spatial 3D lesion coordinates.
- **Attributions & Licenses:** Software: MIT License (see LICENSE). Third-Party Notices: THIRD_PARTY_NOTICES.md. Dataset: UCI Machine Learning Repository (CC BY 4.0). Anatomical 3D Mesh: BodyParts3D © The Database Center for Life Science (CC BY-SA 2.1 JP).